

1. Declaration

I, Ruotong Tang, declare that this assignment, titled *A1.3: Basic App Dev V2 Report*, is my own original work and has not been copied from any other source except where explicitly acknowledged. I have not engaged in plagiarism, collusion, or any other form of academic misconduct in the preparation and submission of this assignment. All sources of information and data used in this assignment have been properly cited and referenced in accordance with the prescribed guidelines. I have not used unauthorized assistance in the preparation of this assignment and have not allowed any other student to copy my work. I am aware that any breach of academic integrity may result in disciplinary action as per the [policies of Monash University](#), which may include failing this assignment or the course, and further academic penalties.

Signature: _____ Ruotong Tang _____

Date: _____ 2025/9/10 _____

2. Github Check

Enter your Github details here.

Github Username <i>Enter your username here</i>	TangRuotong
Repository Shared? <i>Have you started and shared your assignment repository with your tutor yet?</i>	https://github.com/TangRuotong/5032_assignments/tree/BasicDebugExtend

3. Self-Evaluation

Rate your performance for each criteria. Put a ☒ (tick) in the box where you think your work belongs.

Criteria	Exceeds Expectations	Meets Expectations	Needs Improvement	Fail to meet expectations
BR (C.1): Authentication	☒			
BR (C.2): Role-based authentication	☒			
BR (C.3): Rating			☒	
BR (C.4): Security			☒	

4. Screen Recording of BRs

Create a 3 minute video showing your basic web application in action! Upload this video to your Google Drive and put the link here (ensuring that you have updated the access list so its not private).

https://drive.google.com/file/d/15crTiw0jxnNr_YJ6GQL1GQKZYkjmbtr0/view?usp=sharing

5. Reflections: Implementation of C.4 Security

If you have implemented BR C.4, in less than 200 words describe the approach that you have taken to implementing Security in your application. What security flaws were you trying to prevent and what security measures have you implemented to fix those flaws? How do you know that these measures will help prevent those issues from happening? Optionally you can cite external sources to provide evidence for your claim.

To implement BR C.4, the application enforces comprehensive security measures to protect user data and maintain secure interactions. The main security flaws addressed include XSS attacks, weak passwords, unauthorized access, and exposure of sensitive information such as API keys.

To mitigate these risks, the application validates all user input on the client side, including strong password requirements (minimum 6 characters with uppercase, lowercase, numeric, and special characters) and proper email formatting. Role-based authentication ensures users access only permitted pages, preventing unauthorized access to admin functionalities. API keys are stored only in Firebase configuration files on the server side and never exposed directly in client code. Additionally, sensitive data such as passwords is handled through Firebase Authentication, which securely hashes credentials and manages sessions.

These measures reduce the likelihood of XSS attacks, brute-force login attempts, and improper privilege escalation. Client-side validation improves user input integrity, while Firebase's built-in security mechanisms provide robust protection.

6. Reflections: Challenges

What has been the most challenging part of this assignment for you? How has this stretched you as a programmer?

The most challenging part of this assignment was implementing the rating system and securing the application against XSS attacks. Both areas required me to go beyond what I had learned in class and research independently. For the rating feature, I had to figure out how to store user ratings, prevent multiple submissions from the same user, and calculate aggregated scores efficiently. For XSS prevention, I needed to understand potential attack vectors and how to properly sanitize and validate user input to protect the application.

These challenges stretched me as a programmer by forcing me to learn new concepts and tools, such as reactive state management in Vue for the ratings and security best practices from Firebase and OWASP for XSS protection. I also gained experience in thinking critically about user input and potential vulnerabilities,

which strengthened my ability to write both functional and secure code.

7. Declaration: Additional Help

Any tools that you used (including Gen AI or existing code reuse) must be declared here.

Note: GenAI is not allowed for coding purposes in any assignment,

However, you may use GenAI for brainstorming and problem solving. You need to declare all such uses here.
One row per help used.

Name	Description
ChatGPT for debugging help	I used ChatGPT to identify potential causes of bugs and receive guidance on how to fix them when errors occurred.